The Limitations of Generative AI

What these systems cannot (yet) do, and why

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Thesis

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- Public understanding rests on demonstrations of success; the structure of failures is the missing half.
- Failures have structure — **three families**, and the family determines the remedy: **statistical** (sampling-error-like; better data and better measurement, at best), **dynamical** (learning while acting circulates errors through feedback), **normative** (rules, not probabilities — a system trained only to imitate text cannot represent the rule it breaks).
- Outside the models: **hosted delivery** carries measurable research and sovereignty costs, and **incentives** shape the claims as systematically as the builds.

Method
The rest of the talk is the evidence for these sentences; the closing frame audits each one. Claims are marked **observed**, **explained**, or **narrated**.

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This is the deck's abstract, mirroring the paper's: the complete claim set inside ninety seconds, so an early cutoff still leaves the audience with the thesis and its structure. Deliver slowly, and do not argue any sentence yet — each gets its own evidence section. Naming the observed/explained/narrated discipline here puts the labels in place before any number arrives. If the slot collapses to five minutes: title, this slide, the closing audit — the talk still stands.

The record

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What was measured

- All **51 bibliography entries** verified against DBLP, the arXiv API, and Crossref by a rerunnable script; open-access copies of **29** cited papers archived with pinned hashes.
- One claim failed verification and was dropped, not cited — the discipline is applied to this talk's own assertions.
- Quantitative claims were extracted from the archived copies, not from abstracts recalled from memory.

The measured findings ahead
Benchmark overfitting up to **8 points**; style-tuning worth up to **112%** on a downstream benchmark; **205 of 243** public leaderboard models silently deprecated.

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Establish the evidence contract before any claim is made. The verification statement is not decoration: one claim in the paper's own draft failed sourcing and was dropped, which is the discipline applied to ourselves. The three numbers in the block are from the archived PDFs and recur through the talk. Timing 1.5 min. If asked what was dropped: a 2015 surveillance report that could not be re-derived from any reachable source; recorded in VERIFICATION.md.

The failures fall into three families

Family	Signature	Matching remedy
Statistical	errors average out over samples	better data, better measurement
Dynamical	errors circulate through feedback	separate learning from acting
Normative	the violated rule cannot even be represented	inspectable structure outside the weights

The category error to carry home
Offering a statistical remedy for a dynamical or normative failure — the most common shape of announced fixes and AI policy debates alike.

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This table organizes the evidence that follows; everything ahead fills in one row. Walk left to right: family, signature, matching remedy. The category error is the practical takeaway: a concrete hook is “more data” offered as the remedy for a privacy violation — a statistical remedy for a normative failure. One diagnostic question for any announced fix: which family does the failure belong to? 2 min.

Evidence: the failure families

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Confabulation is the training objective working as designed

- "Hallucination" suggests a perception problem; the mechanism is the objective itself rewarding plausible-sounding continuation.
- Observed: scaling *dilutes* but does not remove it; retrieval patches the symptom on popular topics while the long tail stays exposed (Mallen et al., 2023).
- The state is *malleable*: a masked-word test that changes one word of the prompt flips the answer it gives (Pride & Prejudice marriage dates, .org/.com) — what is "known" versus what is reported come apart.

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Set up the reframe precisely: "hallucination" implies a broken perception module, but the mechanism is the objective rewarding plausible continuation — the behaviour follows from the training objective. Mallen et al.: retrieval patches the popular tail; the long tail stays exposed. The malleability result is the most concrete datum: changing one word flips the answer, so held knowledge and reported output come apart. If asked whether scale fixes it: the observed result is dilution, not removal.

Context and calibration

Context is a budget, not a capability

- Effective vs. advertised length: measured gap (Hsieh et al., 2024, RULER); "lost in the middle": recall is high at the edges and collapses in between (Liu et al., 2024).
- Retrieval augmentation inherits the problem: whatever is retrieved poorly is silently absent.

Calibration

- A system that is right 80% of the time should say so; badly calibrated systems sound equally sure either way.
- Masked-word experiments show the disconnect between held knowledge and reported confidence directly.

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Two distinct failures share a slide — keep them separate. Context: RULER measured the gap between effective and advertised length; Liu's "lost in the middle" is the positional curve — edges good, middle collapsed. The retrieval point matters for deployed systems: whatever retrieval misses is silently absent, and the user cannot see the hole. Calibration: right 80% of the time should sound like 80%; poor calibration means equal confidence on correct and incorrect answers. Bridge: "and the tests we measure this with are themselves unreliable" — next slide.

Even the measurement layer is compromised

- **Contamination**: test questions leak into training data — examinations become memory tests.
- Fresh GSM1k problems matching the GSM8k distribution: drops of up to 8 percentage points in several model families — though, for honesty's sake, the frontier systems showed little of it (Zhang et al., 2024).
- So a headline benchmark number can conflate reasoning with memorization, and you cannot tell which from the score alone.

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Contamination first: leaked test questions turn examinations into memory tests. GSM1k is the headline: fresh problems, same distribution, up to 8 points down. State the honesty clause without prompting: the frontier systems showed little drop — report the result that weakens the dramatic reading. The conclusion is careful: the score alone cannot distinguish reasoning from memorization, so the headline number is itself the unreliable artifact. If challenged on GSM1k's validity: it matches GSM8k's distribution by construction; the matching procedure is documented in the paper.

Memory: the read side of state

- A fixed context window meets unbounded interaction: long sessions degrade retrieval of their own beginnings — forgetting what was said *in this conversation*.
- This is architectural, not a data problem: no amount of better training data changes the shape.
- Remedies that match the mechanism: explicit external state — scratchpads, retrieval over interaction history, persistent stores — with write policies that are themselves inspectable.

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Transition from errors that average out to errors that persist by construction: a fixed window meets unbounded interaction, and the session loses access to its own beginnings. Emphasize: architectural, not a data problem — no training corpus changes the shape of the window. The remedy side matters for later questions: external state with inspectable write policies. Audience recognition helps here — the lived experience of repeating oneself by turn forty confirms the mechanism.

Learning in the loop: when errors circulate

- Updating a network inside its own control loop is a feedback system, with the failure modes feedback systems have:
  - bootstrapped updates diverge — provably, even with linear approximation (Baird 1995; Tsitsiklis & Van Roy 1997)
  - agents overfit their earliest experience (Nikishin et al., 2022)
  - shifted data overwrites prior learning (French, 1999)
  - prolonged continual training destroys the capacity to learn at all (Dohare et al., 2024)
- The remedy is **architectural separation**: learning on one timescale, acting on another (Borkar, 1997; Feldbaum's dual control).
- Statistical remedies (more data, better prompts) do not touch this family.

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Keep this precise — the section's authority is that the failures are theorems and replicated phenomena, not intuitions. The four cited results: divergence proofs, primacy bias, catastrophic forgetting, plasticity loss. The remedy is the classical one from adaptive control: two-timescale separation of estimation from action. Close the section on the table callback: statistical remedies do not touch this family. That sentence is the section's takeaway.

Some obligations are rules, not probabilities

- Privacy (contextual integrity, Nissenbaum 2009) is a system of norms: who may transmit what about whom, to whom, in which context.
- Formalized, it becomes a deontic logic with temporal operators (Barth et al., 2006) — requiring recipient beliefs, role and group structure, retention over time, leak probabilities.
- Read the machinery off the formalization: multi-type, multi-agent, multimodal logic with temporality and probabilism.
- Nothing in a corpus gradient can represent “this transmission violates this norm in this context.”

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The pivot from measured behaviour to representational limits — slow down. Privacy via Nissenbaum’s contextual integrity: who may transmit what about whom, to whom, in which context. Then the formal move: it becomes a deontic logic with temporal operators (Barth et al.). The machinery requirement is read off the formalization, not asserted: multi-type, multi-agent, multimodal, temporal, probabilistic. The closing sentence is the section’s claim: nothing in a corpus gradient can represent “this transmission violates this norm in this context.” If someone raises refusals, defer explicitly: next slide.

Instances are not coherence

- A fine-tuned model sometimes refuses to repeat an address: unreliably, without audit trail or consistency, correctable only by retraining.
- Coherence — norm consistency, auditability, reversibility — requires **inspectable structure outside the trained weights**.
- The semiotic layer matters: an *indexical* link (this photo *is* this person) is a doxxing vector; an iconic rendering, representational harm; a symbolic misuse violates an identifiable norm.

Deployment reality
Narrow hybrids (guardrails, constrained decoding) are commercial; flagship neurosymbolic systems (AlphaGeometry, AlphaProof) are research stage; the full governance-demanding profile is deployed nowhere — the vacuum is the finding.

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Answer the deferred objection directly: yes, fine-tuned models sometimes refuse — unreliably, with no audit trail, no consistency, and no correction short of retraining. Instances are not coherence. The semiotic aside works for philosophically inclined audiences: an indexical link (this photo is this person) is a doxxing vector; an iconic rendering is representational harm; a symbolic misuse violates an identifiable norm. The deployment block is observational, not rhetorical: narrow hybrids are commercial, flagship neurosymbolic is research-stage, the full profile is deployed nowhere — and that vacuum is itself a finding about the field.

Evidence: the alternatives

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Diffusion language models change the trade-offs

- Generate all positions in parallel, in any order; every token is conditioned on every other (bidirectional context).
- Observed: an 8B diffusion LM is competitive with an equivalently sized autoregressive model (Nie et al., 2025, LLaDA vs. LLaMA 3 8B).
- Caveats: unlearning pathological associations is harder when all tokens are mutually conditioned; latency wins come at scale.
- Scope: helps *statistical* limits (all-position access for context, parallel generation) — not the dynamical or normative families.

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State the scope first or this slide will be misheard as "diffusion fixes everything." What diffusion changes: all positions in parallel, bidirectional conditioning; LLaDA 8B is competitive with LLaMA 3 8B (Nie et al., 2025). The caveats are part of the record: unlearning is harder under mutual conditioning, and latency wins materialize at scale. Scope line: this addresses the statistical family only — it remains a distribution learner; no deontic structure appears. This slide exists so the alternatives section is not left unexplored.

Evidence: the delivery layer

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The shape of the machine shapes the science

- Weights withheld, service only: transparency scores are lowest exactly where a service model hides things — training data, compute, hardware (Bommasani et al., 2023, FMTI).
- Hosted models are updated or retired without documentation, and published findings silently break (Pozzobon et al., 2023).
- Access to scale was concentrating before the boom (Ahmed, 2020); ML-based science has a reproducibility crisis of its own (Kapoor & Narayanan, 2023).
- An internal Google document (2023) acknowledged the moat reasoning directly; competing explanations (safety, legal exposure) predict similar behaviour, so the *motive* stays open — the measurable cost to the research commons needs no hedge.

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Section pivot: from what the systems cannot do to how they are delivered and studied. FMTI: transparency is lowest exactly where the service model hides things (data, compute, hardware). Pozzobon: hosted models updated or retired without documentation — published findings silently break, which is a reproducibility problem for the field. Ahmed: concentration predates the boom; Kapoor & Narayanan: ML-based science has its own leakage-and-reproducibility crisis. The moat bullet is delivered with the paper's exact hedge: the document acknowledged the reasoning, competing motives predict the same behaviour, the motive stays open — the measurable damage to the research commons is what can be stated without hedging.

Why states now treat hosted AI as a dependency question

- A hosted service runs on someone else's infrastructure and jurisdiction, and the history keeps arriving: Merkel's phone (2013), Crypto AG secretly CIA-owned (2020), Danish cables used against European politicians (2021), a nine-year campaign of surveillance and intimidation against ICC staff (2024), ECHELON, and *Schrems II* (2020).
- The lesson generalizes: exposure is a property of who operates the infrastructure, not of who holds which alliance today.
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- Hence *digitale Souveränität*: procurement moves (Schleswig-Holstein, Denmark) and the Data Act's switching and portability obligations.

Keep this factual and unhurried — the incident trail does the work, and it is deliberately multi-actor: Merkel's phone (2013), Crypto AG secretly CIA-owned (2020), Danish cables used against European politicians (2021), the nine-year surveillance-and-intimidation campaign against ICC staff (2024), ECHELON, Schrems II. The generalization is the section's key sentence: exposure is a property of who operates the infrastructure, not of who holds which alliance today — which is why the incident set implicates allies and adversaries alike. Close on digitale Souveränität: Schleswig-Holstein and Denmark procurement moves, the Data Act's switching and portability obligations. 2 min.

The everyday side: what does the customer possess?

- A self-contained product keeps working when its maker does not; a service that shuts down — acquisition, repricing, lost interest, bankruptcy — takes its function with it.
- Not distinctive to AI: the standard lifecycle risk of hosting — which is exactly what the sovereignty programs legislate against.
- Self-hosting and open weights exist in the design space (the Mowgli stack is a small, deliberately modest existence proof); the question is why the dominant presentation is the one where the customer owns least.

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The everyday counterweight to the geopolitical slide: what does the customer possess? A self-contained product outlives its maker; a service shutdown — acquisition, repricing, lost interest, bankruptcy — takes the function with it. This is not distinctive to AI; it is the standard lifecycle risk of hosting, which is what the sovereignty programs legislate switching and portability against. Name Mowgli as a deliberately modest existence proof, not a competitor. The closing question — why the dominant presentation is the one where the customer owns least — is held open for the incentives slides.

Misrepresentation needs no dishonesty

- Each link is individually reasonable; the chain is not: vendors fund demonstrations of success, the visible scoreboard is run with the firms it ranks, and the products are tuned for approval.
- **Sycophancy** (Sharma et al., 2023): agreement with the user's stated belief across every assistant studied, traced to human preference data and amplified by preference-model optimization — apparent agreement is not evidence of correctness.
- **Leaderboard Illusion** (Singh et al., 2025): up to 27 private variants tested in parallel by select providers; 205 of 243 public models silently deprecated; restricted tuning worth up to 112% on a downstream benchmark.

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The section's central sentence, and the one that keeps the paper defensible: misrepresentation needs no dishonesty — each link is individually reasonable; the chain is not. Sycophancy (Sharma): agreement with stated belief across every assistant studied, traced to preference data and amplified by preference-model optimization — so apparent agreement is not evidence of correctness. Leaderboard Illusion (Singh) carries the numbers, delivered slowly: up to 27 private variants tested in parallel by select providers; 205 of 243 public models silently deprecated; restricted tuning worth up to 112% on a downstream benchmark. Let the numbers sit without commentary.

Honesty about the boundary

Measured
The capability claims are shaped by the same incentives that selected the architecture: this half is evidenced.

Open, falsifiable
Whether those incentives *reduce* total progress or merely *redistribute* it toward the measurable and saleable. *Weakening conditions*: independent instruments anchoring public perception; replication failure of the asymmetries; their absence in deployed systems.

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The boundary slide — deliver it with precision, this is where speakers overreach. Measured: capability claims are shaped by the same incentives that selected the architecture. Open and falsifiable: whether those incentives reduce total progress or merely redistribute it toward the measurable and the saleable. Read the weakening conditions aloud: independent instruments anchoring perception, replication failure of the asymmetries, their absence in deployed systems. It is worth saying explicitly: this is the claim most worth being wrong about, and the conditions under which the audience would show it are stated. That sentence is the talk's credibility in miniature.

Discussion

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What remains open

- Is confabulation reducible to acceptable levels under an objective that rewards plausibility? (*unresolved*)
- Do effective and advertised context lengths converge? (*persistent across generations*)
- Is reliable compositionality reachable by training alone? (*evidence points to structural support*)
- Do the incentives slow progress, or redistribute it? (*falsifiable; conditions stated*)

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In-principle limits and in-practice shortfalls are different claims, and conflating them is how both hype and dismissal manufacture their certainty.

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Recap the open problems with their status tags — unresolved, persistent across generations, structural support pointed to, falsifiable with stated conditions. Overclaiming none of them is the point; the tags are the content. The closing block is the talk's second-most-quotable sentence: in-principle limits and in-practice shortfalls are different claims, and conflating them is how both hype and dismissal manufacture their certainty. Pause after it.

Takeaways

1. Ask **which family** of failure you are looking at — the remedy only exists in the matching family.
2. Distrust single-number evaluations: fresh test sets, independent instruments, style-sensitivity checks.
3. For deployed decisions, ask what you **possess**: weights, audit rights, switching paths — or only a subscription.
4. Coherence (privacy, audit, consistency) demands inspectable structure; a probability distribution cannot break a rule it cannot represent.

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Four takeaways, one breath each: (1) ask which family of failure — the remedy exists only in the matching family; (2) distrust single-number evaluations — fresh test sets, independent instruments, style-sensitivity checks; (3) for deployed decisions ask what you possess — weights, audit rights, switching paths, or only a subscription; (4) coherence demands inspectable structure — a probability distribution cannot break a rule it cannot represent. If time is short, cut nothing here — cut the diffusion slide and compress the open questions. These four are what the audience should be able to reproduce tomorrow.

The thesis, audited

- **Family determines remedy** — the taxonomy; fixes fail by mismatch (*explained*).
- **Statistical: data and measurement, at best** — dilution-not-removal (Mallen); RULER's gap; GSM1k's 8 points, frontier excepted.
- **Dynamical: errors circulate** — divergence proofs (Baird 1995; Tsitsiklis & Van Roy 1997); primacy bias (2022); plasticity loss (Dohare 2024).
- **Normative: rules need representation** — Barth et al. (2006); deployed nowhere: the vacuum is the finding.
- **Delivery costs are measurable** — FMTI (2023); Pozzobon (2023); the incident trail; the Data Act's switching duties.
- **Incentives shape claims** — sycophancy across every assistant studied (Sharma 2023); 27 variants, 205 of 243, 112% (Singh 2025).
- **Open, falsifiable**: reduce or redistribute progress — weakening conditions on the boundary slide.

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- Incentives shape claims — sycophancy across every assistant studied (Sharma 2023); 27 variants, 205 of 243, 112% (Singh 2025).
- Open, falsifiable: reduce or redistribute progress — weakening conditions on the boundary slide.

The audit closes the pyramid: each thesis sentence with the evidence that supports it, named precisely enough to be checked — citation, year, number. If any thesis sentence is challenged in questions, this frame is the map to the answer. The one sentence not closed here — reduce versus redistribute — is the boundary slide's falsifiable remainder; point at its weakening conditions rather than arguing it. Takeaways have already been shown; hold this frame and then the closing URLs for questions.

The Limitations of Generative AI

- Paper (with verified bibliography and corrections log): `https://nadiayvette.frama.io/gen-ai-limits/main.pdf`
- Source on all mirrors; framagit is canonical: `https://framagit.org/NadiaYvette/gen-ai-limits`

Thank you.

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Thank you.

Leave this up during questions — the URLs are the call to action. Expected question clusters and one-line responses: (1) “So AI is useless?” — no; the claim is that failure structure is knowable and remedies are family-specific. (2) “Isn’t neurosymbolic too hard or slow?” — deployed-nowhere is an observation about the field’s current state, not an argument that it must remain so. (3) “Isn’t this just anti-industry?” — the observed/explained/narrated labels apply to the critique as well; the falsifiability slide states the weakening conditions. (4) On any specific number: offer the bibliography — every figure on these slides traces to a hash-pinned PDF.